

Field Guide

A student-led campus-forest PWA for Ateneo Loyola Heights — walk the campus, meet a tree, log what you saw.

LIVE

yclap-field-guide.marangelonrevelo.workers.dev

SURFACES

Home · Map · Journal · Plan

STATUS

Deployed · 65/65 tests green

AS OF

2 Sep 2026

What it is

A phone-first field guide to the trees on our own campus. You open it on a walk, it tells you what is near you, you photograph a tree, iNaturalist suggests what it is, and it goes into *your* journal. It works with no signal.

1,809

Trees on campus

AIS figure, SY 2025–26. Not our survey.

101

In the arboretum

AIS figure. Cited on screen.

9

Species curated

Each fact carries a source.

186

Walkable paths

OSM footways inside the campus box.

The four surfaces

/

Home

The campus in three sourced numbers, the species you can meet, and the one gap said out loud: AIS has a species database, but not a count and location per tree.

/map

Map

Real satellite or street tiles, your position projected through Web Mercator, encounters that fire at 25 m and stop hedging at 8 m. Start a walk, open the camera, log a sighting.

/journal

Journal

Your collection. Every sighting keeps species, photo, note, walk, coordinate, accuracy — and whether the fix was real or the demo loop. Exports to GeoJSON and CSV.

/plan

Plan

What we are asking for and from whom — the AIS count-and-location gap, CFMO grounds data, Ateneo Wild's photo catalogue. The asks are on the record, not in a slide.

What ships today

Every row below is running in the deployed build.

Map without a map library	A slippery map written here in ~270 lines — pan, zoom about the cursor, real Web Mercator. Four key-free basemaps: Guide, Trail, Paper, Satellite. Each was probed at a campus tile before it was allowed to ship; three candidates were rejected for watermarks, missing zoom levels, or a 404.
Position you can trust	<code>watchPosition</code> projected through the same projection the tile server drew with, so the marker stays on its feature at every zoom. The accuracy ring is a real radius in real metres.
Demo campus	A scripted 42-second walk over the real paths, on by default — a hall off Katipunan has no useful GPS fix, and a stage demo must not depend on one. Toggle it off to use the device.
Encounters	True haversine metres. Within 25 m an encounter pins and gets an arrival ring; within 8 m the card stops hedging. It follows the nearest tree until you tap one.
Camera	Live rear-facing viewfinder → canvas → JPEG at ≤1024 px. Falls back to a file picker when permission is denied or there is no camera. The stream is torn down on capture.
Identification	The photo is POSTed to iNaturalist's computer vision. This app runs no model and the caption says so. With no token it replays a recorded reply and announces that in capitals.
Walk sessions	Start and end a walk; every sighting saved inside it carries a walk id, coordinate, accuracy, and whether the fix was real.
Journal & export	<code>localStorage</code> — no account, no server, no other student's data. Summary is counts and groupings. Exports located sightings as GeoJSON and CSV; photos are deliberately excluded.
Offline	A hand-written service worker precaches the shell and cache-firsts assets and map tiles, capped at 1400 with oldest-first eviction. Tap "Save offline" once before you go on stage. Verified 2 Sep against a harness that blocks the service worker too, not just the page.

How it is built

Stack. React and react-dom, and nothing else. The slippery map and the service worker are hand-written — handing both to a map library to get pan, zoom and correctly-placed markers was the worse trade for a 300 KB bundle that has to work offline.

Verified. 65 tests across 21 suites, all green. Type-check clean. Offline proven against a harness that emulates the network on the service worker too — the naive version blocks the page and lets every tile through, which proves nothing.

Deployed. Cloudflare Workers with static assets, SPA fallback so `/map` survives a reload. HTTPS is not cosmetic here — the camera will not open without it.

Before a demo. Tap *Save offline* once on the basemap you plan to show. Leave *Demo campus* on. Add an iNaturalist token if you want live identification — it expires in about a day.

How we make it play

The app is built on a specific, defensible position about game mechanics — one drawn from Ateneo's own published research, and one that is enforced by an automated test rather than by good intentions.

Counting is not a breach of the rule.

Comparing is.

The line every engagement feature in this app is held to. Derived from a survey of 40+ nature-app interfaces: not one of them ranks its users against each other. Every "progress" number in the reference set — *3 of 5 discoveries, 47 species, 2.8 km · 46 min · 7 species* — is about *this walk or my collection*.

So the app is a **collection game**, not a competition. You are playing against the campus and against your own last walk — never against another student. Two sources put us here: Nicholson's *Meaningful Gamification*, and RECIPE (Rodrigo, Favis & Cuyegkeng 2021), the framework behind Ateneo's own SDG game. The journal carries their line on screen: "**Reflection, not a race.**"

PLAYING TODAY

- shipped** **A collection to fill.** Nine curated species as a grid of badges, with empty slots for what you have not met yet.
- shipped** **Discovery by proximity.** Trees announce themselves as you get near — 25 m to appear, 8 m to confirm. The walk is the input device.
- shipped** **The capture moment.** One big round shutter, a live viewfinder, then a species suggestion. This is the loop people repeat.
- shipped** **A session with a boundary.** Start walk / end walk gives play a beginning and an end, and tags everything caught inside it.
- shipped** **Your own record.** Sightings, species, located count, by-species bars, by-day pills — all yours, all local.

PROPOSED BEFORE 12 SEP

- open** **A walk with a shape.** Ordered stops and a progress rail reading *n of N*, resetting each walk.
- open** **A receipt, not a stop.** Ending a walk shows distance, elapsed time, species seen, and which were *new to you*.
- open** **Seen-of-total.** The journal header reads *4 of 9 species seen*, and each sighting gets a stable specimen index.
- open** **A question, not just a form.** Saving a sighting asks what you noticed — optional, never blocking the save.
- open** **Report a tree we do not have.** Turns the AIS gap into something a student can act on, and it leaves in the export.

The honesty invariants

Rules that must survive any edit to this app. They are the reason a judge can trust the rest of it.

- ✓ **AIS's numbers are AIS's.** 1,809 trees, 101 in the arboretum, ~2% green cover — each carries a source and a year on screen. Nothing is labelled "our survey."
- ✓ **The AIS gap is said out loud.** AIS has the species database; what it lacks is count and location per tree. That sentence appears on Home, Map and Plan.
- ✓ **iNaturalist identifies, not this app.** No model weights ship here, and with no token the app says its reply was recorded — it never claims to have looked at your photo.
- ✓ **Encounter coordinates are positions on a demo map,** not surveyed tree locations.
- ✓ **The paths and campus outline are OpenStreetMap's,** credited on the map whenever the layer is on. Not a survey, not a cadastral boundary.
- ✓ **Restricted ground stays hatched** and spawns no encounters. The scripted demo walk was re-routed when it was found crossing it.
- ✓ **No leaderboard, no points, no rank, no streak** — asserted by test, not by memory.

What we turned down, and why

These are logged refusals with named sources, not oversights — which is why they are worth showing rather than hiding.

REFUSED	REASON ON RECORD
refused Public leaderboard, XP, rank, points-per-photo	Sophie: against incentivising advocacy involvement. Ivan: no prize budget unless a sponsor appears. A prize-driven walk stops when the prize does.
refused Streaks	Punishes the student who misses a day. The journal is a record, not an obligation.
refused Community feed, follows, avatar pins of other students	Turns a walk into an audience. Also a real safeguarding question about broadcasting where students are.
refused CO ₂ -offset / survival-rate impact dashboard	We do not measure it. A number we cannot source is the fastest way to lose a judge.
refused Steps, calories, weather	Same rule — we do not measure them.

A DECISION THE TEAM SHOULD MAKE OUT LOUD

The no-ranking rule is currently enforced by a test: `journal.test.ts` fails the build if a rank, score, points, streak, level or xp field ever appears in the journal summary. That is deliberate — it stops the rule eroding quietly under deadline. If the team now wants competitive mechanics, that is a legitimate call, but it should be made explicitly, with Sophie and Ivan in the room, and the test changed on purpose — not worked around.

The nine species

Curated, not scraped. Every claim on a species card traces to a citation carried in the app.

Narra <i>Pterocarpus indicus</i> Native National tree Shade	Molave <i>Vitex parviflora</i> Native Threatened Hardwood	Katmon <i>Dillenia philippinensis</i> Native Endemic Understory
Dao <i>Dracontomelon dao</i> Native Canopy	Mahogany <i>Swietenia macrophylla</i> Exotic Fast shade	Rain tree <i>Samanea saman</i> Exotic Wide shade
Teak <i>Tectona grandis</i> Exotic Timber	Balete <i>Ficus sp.</i> Native Strangler fig Landmark	Lagundi <i>Vitex negundo</i> Native Shrub Medicinal

WHY LAGUNDI AND MOLAVE STAY APART

Both are *Vitex*, and the two get merged constantly — including on our own early species list. They are different plants, and the app keeps them as separate records with a caution on each card naming the other. Getting a species wrong in public is exactly the kind of error a botanist in the audience will catch.